Using Jupyter Lab

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> **pip Installation Note:** Ubuntu 24.04 marks the system Python as "externally managed" according to PEP 668, so installing directly with pip may report `externally-managed-environment`. `--break-system-packages` allows pip to install packages in that environment; a virtual environment usually does not need this parameter, but this tutorial keeps it to keep the commands consistent.

> The factory system already has the relevant environment set up, so the following installation steps can be skipped.

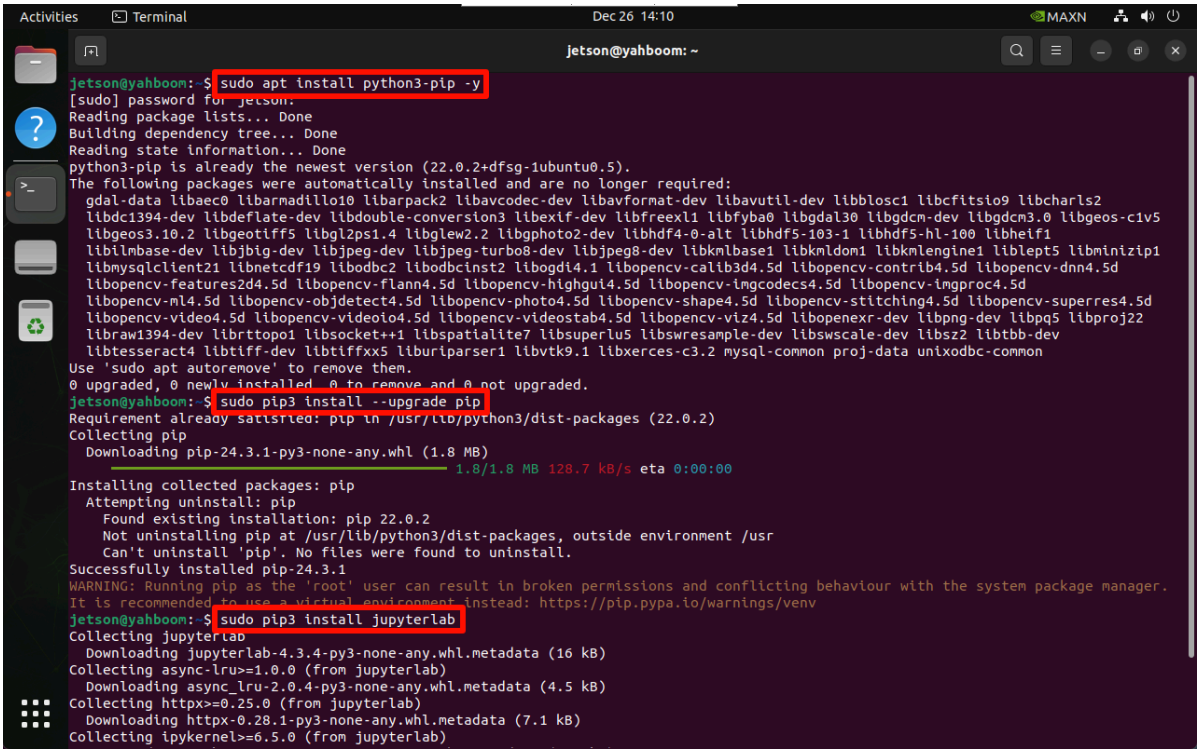
1. Jupyter Lab Installation

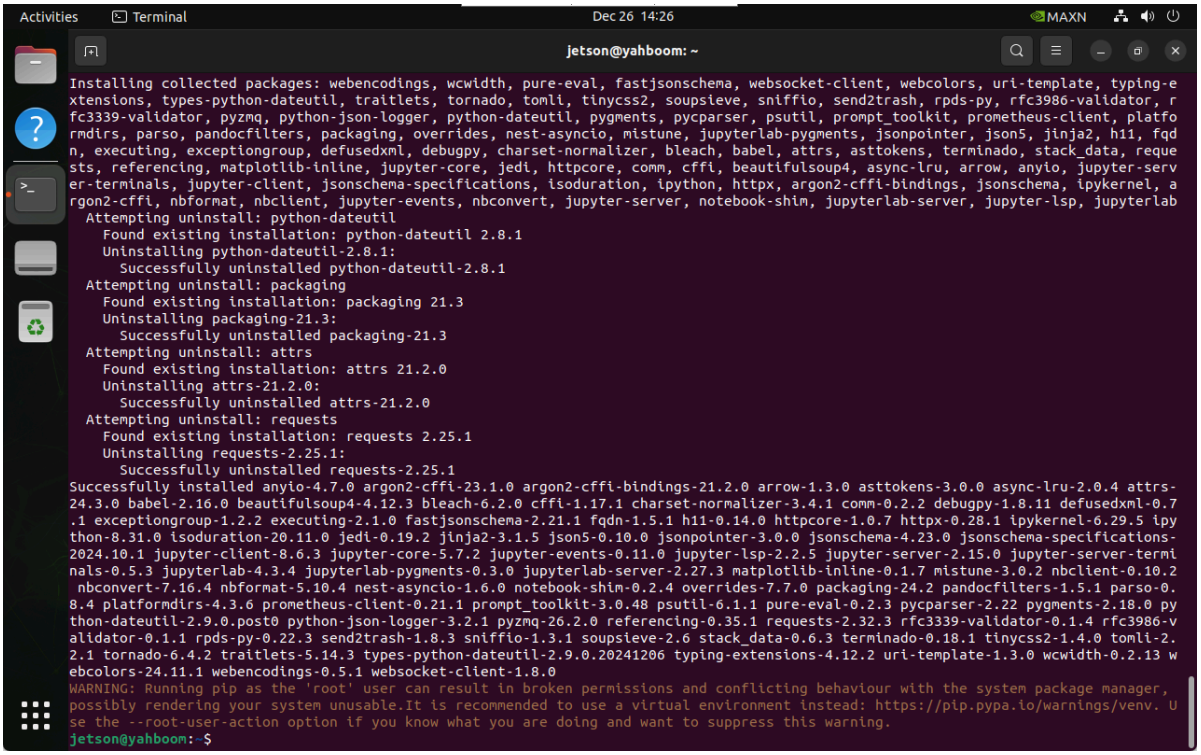
1.1. Jupyter Lab

Use the following commands to install Jupyter Lab: if the download speed for installing Jupyter Lab is slow, you can use a specified source for the installation

```
sudo apt update
sudo apt install python3-pip -y
sudo pip3 install --upgrade pip --break-system-packages
```

```
sudo pip3 install jupyterlab --break-system-packages
# Tsinghua source: pip3 install jupyterlab -i
https://pypi.tuna.tsinghua.edu.cn/simple --break-system-packages
# Alibaba Cloud source: sudo pip3 install jupyterlab -i
https://mirrors.aliyun.com/pypi/simple/ --break-system-packages
```





1.2. Node.js

Use the following commands to install the latest Node.js:

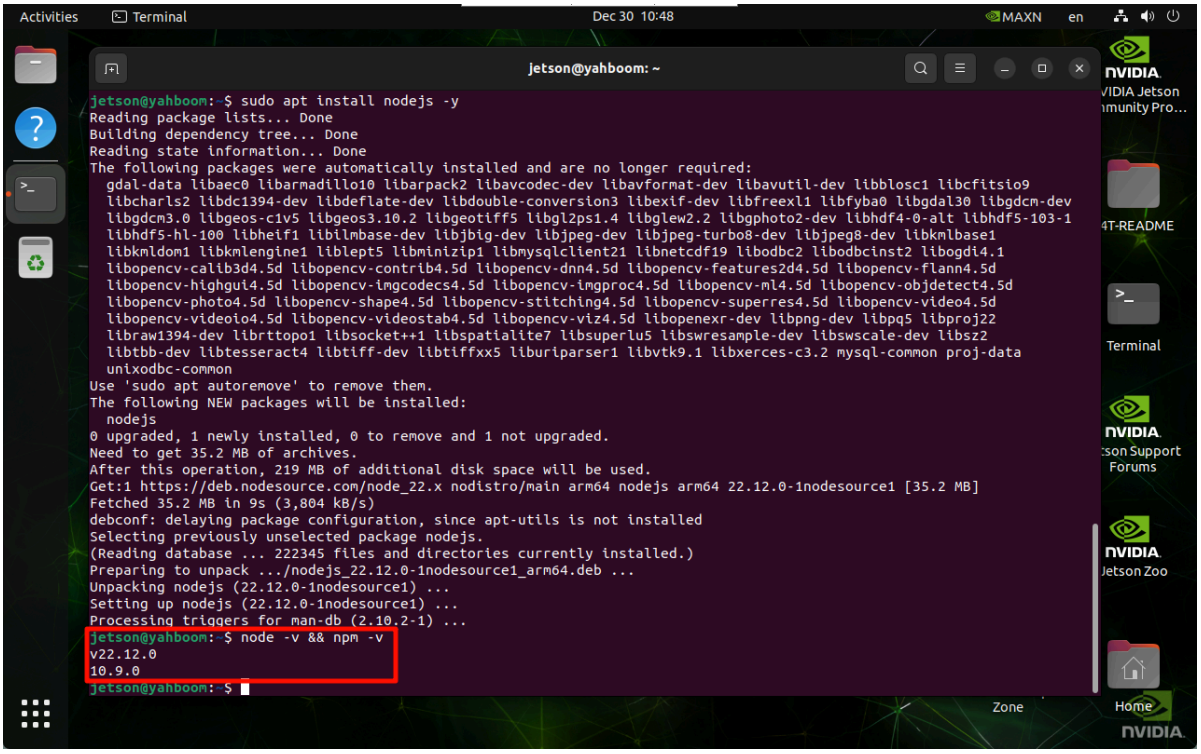
```
sudo apt install curl -y
```

```
sudo curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
```

```
sudo apt install nodejs -y
```

Verify the version:

```
node -v && npm -v
```

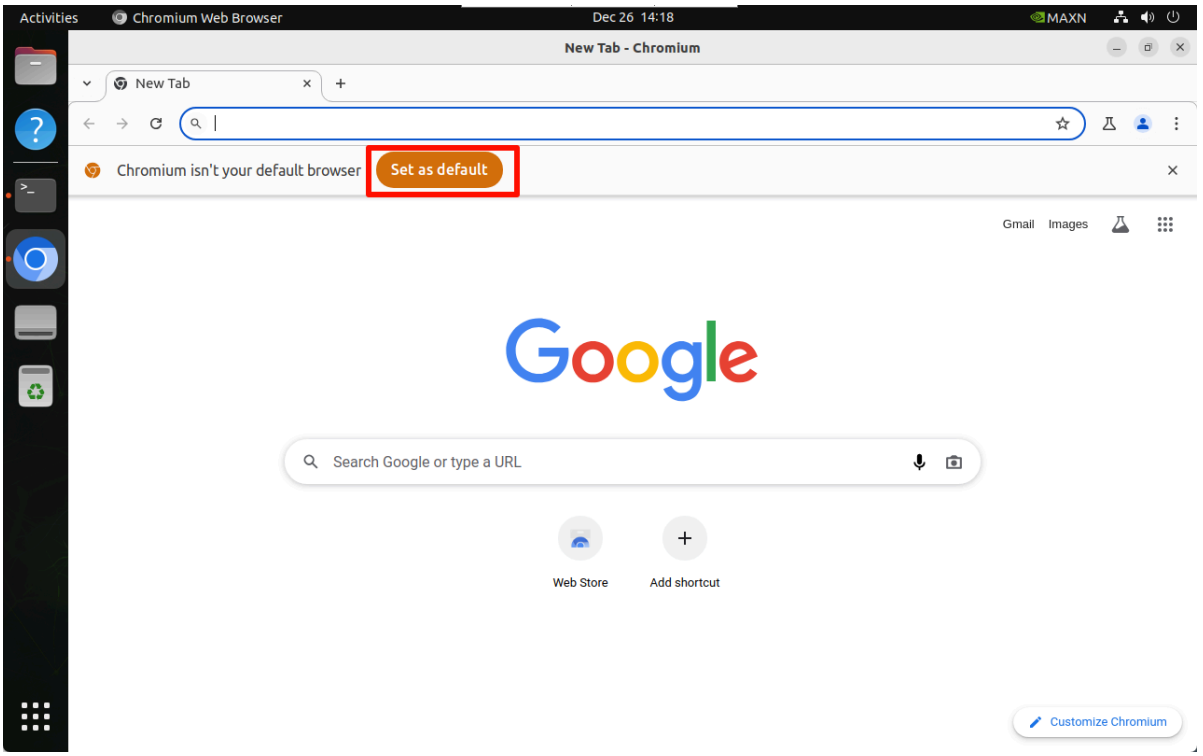


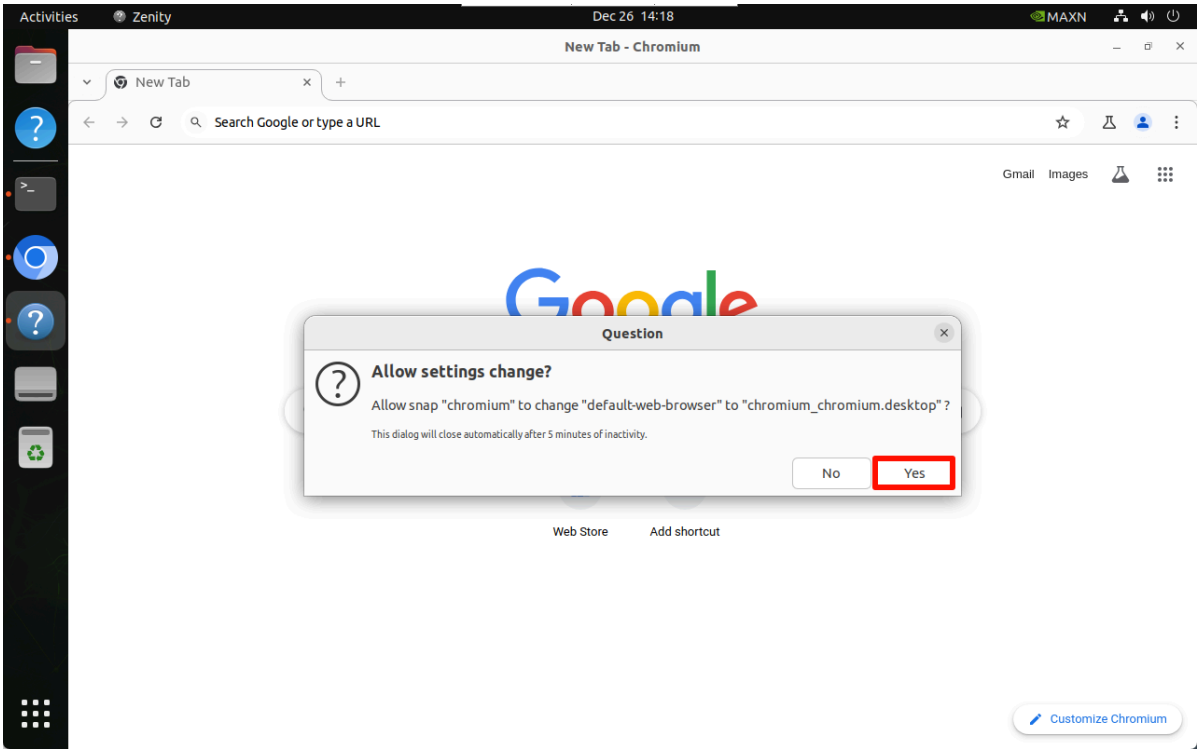
2. Jupyter Lab Startup

Before starting Jupyter Lab, the system default browser needs to be set; otherwise, some prompts appear when the terminal is started.

2.1. Set the Default Browser

Open the system Chromium browser and choose to set it as the default browser:

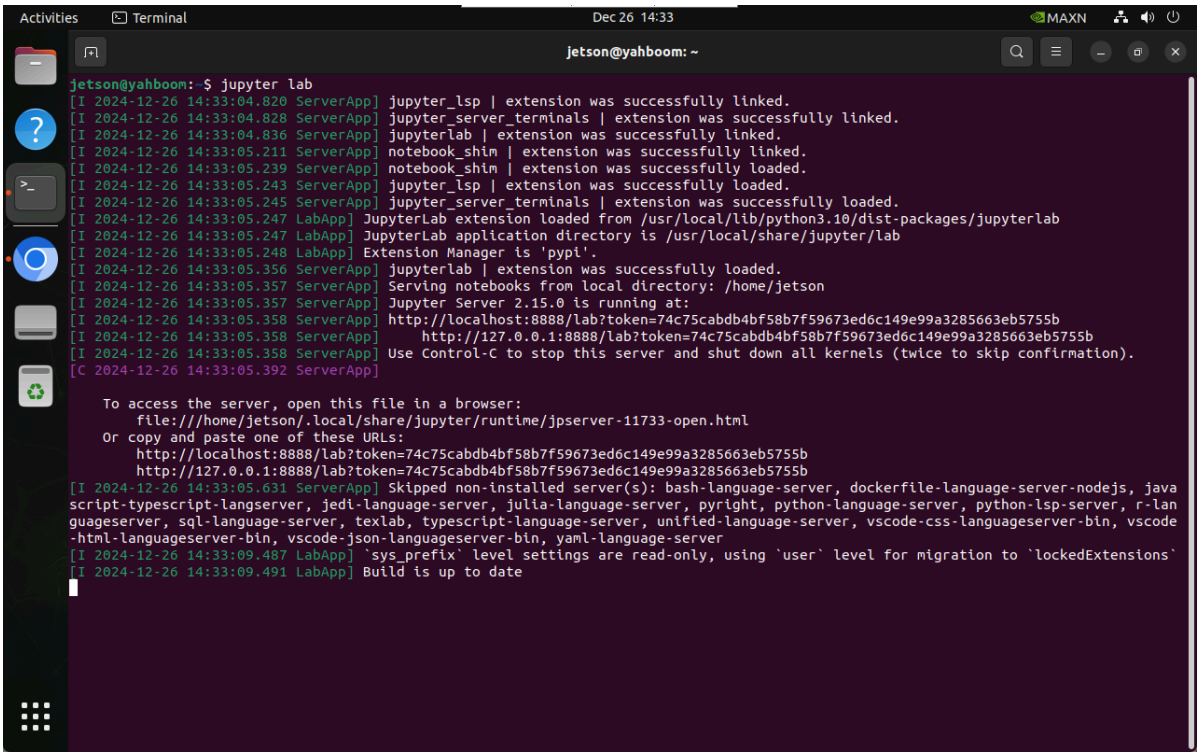


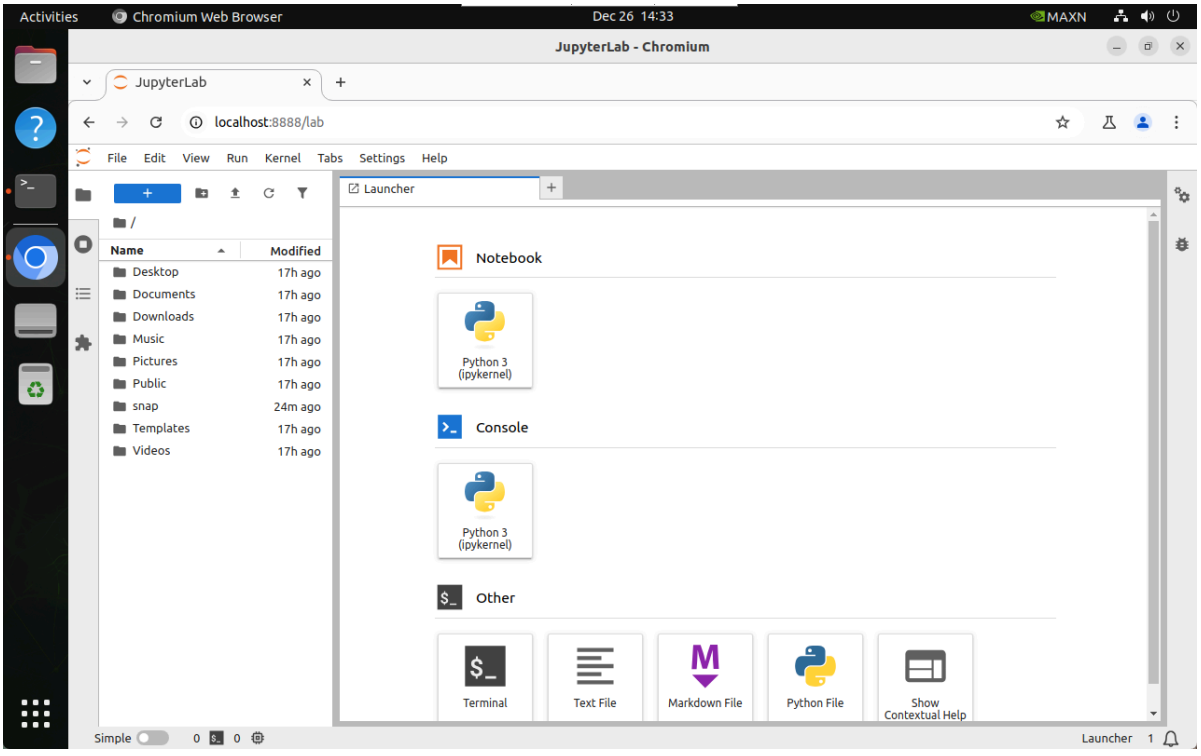


2.2. Start Jupyter Lab

> The factory system has been set to start Jupyterlab automatically at boot, so you can jump directly to 2.3 to check.

```
jupyter lab
# Start without a browser jupyter lab --no-browser
# Start as administrator sudo jupyter lab --allow-root
```

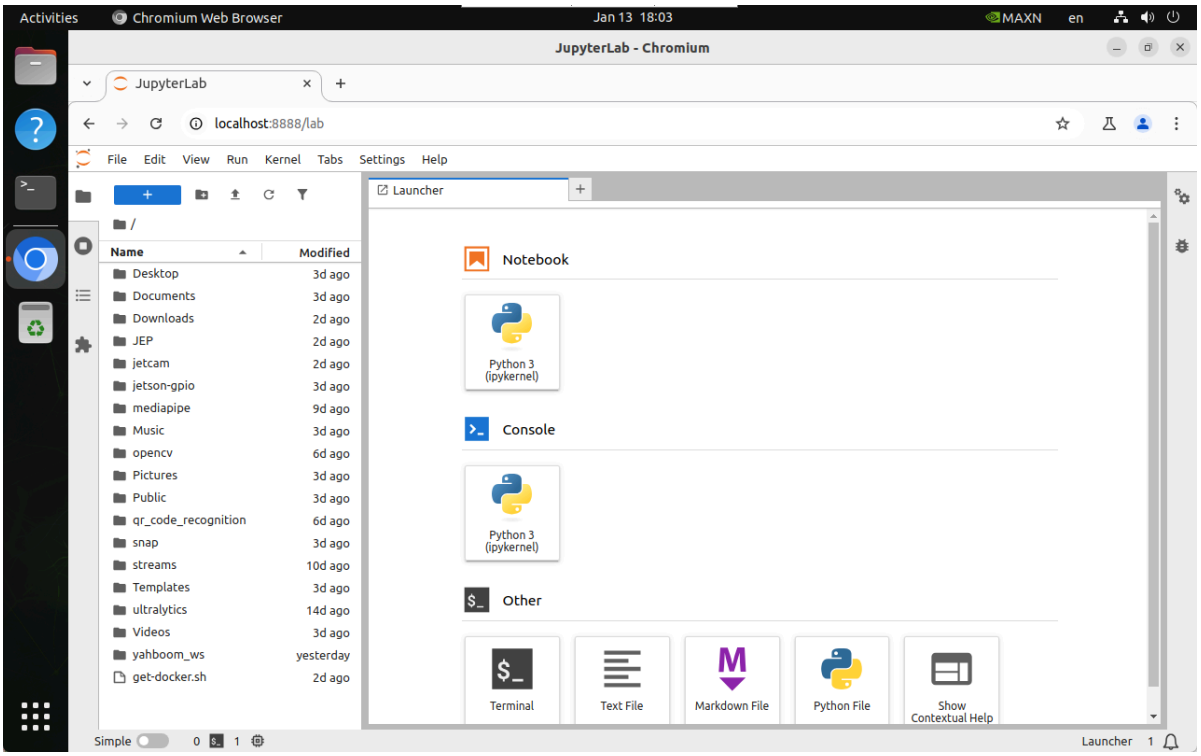




2.3. Host Access

The host refers to access from the Jetson mainboard system; access it directly through http://localhost:8888/:

```
http://localhost:8888/
```



3. Jupyter Lab Configuration

Configure LAN access, the access password, auto-start at boot, and other operations for Jupyter Lab.

3.1. LAN Access

Set it so that devices on the same LAN can enter IP:8888 in a browser for access!

Note: The LAN of a campus network is generally inaccessible; you can switch to a laptop/mobile phone hotspot for testing

```
For example, the mainboard IP is 192.168.2.114; you can enter 192.168.2.114:8888 in a browser on the same LAN to access the Jupyter Lab on the mainboard
```

3.1.1. Create the Configuration File

```
sudo jupyter lab --generate-config
```

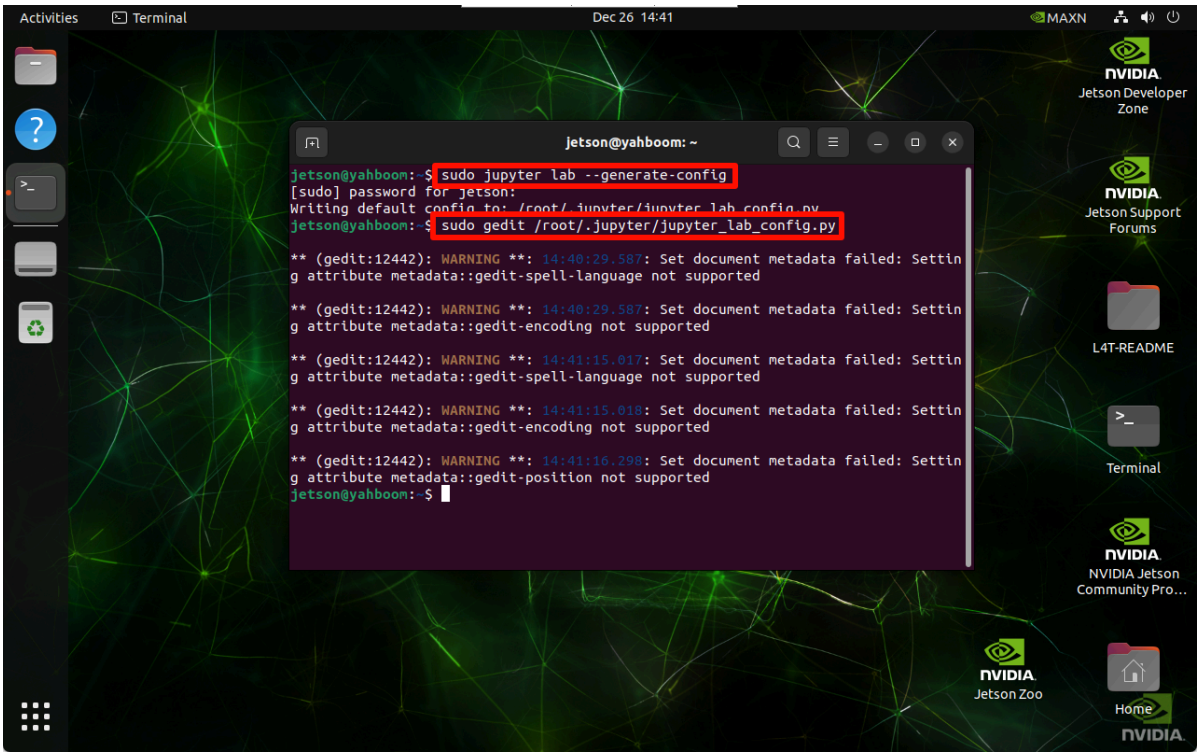
Location of the automatically generated configuration file: Writing default config to: /root/.jupyter/jupyter_lab_config.py

3.1.2. Modify the Configuration File

```
sudo gedit /root/.jupyter/jupyter_lab_config.py
```

Modified content: after modifying, click save and close the file

```
# Allow requests from any source to access the Jupyter Lab server
c.ServerApp.allow_origin = '*'
# 0.0.0.0 means binding all available network interfaces, allowing access from
any address
c.ServerApp.ip = '0.0.0.0'
# Allow starting the Jupyter Lab server as the root user
c.ServerApp.allow_root = True
# Modify the default port to avoid conflicts
c.ServerApp.port = 8888
```

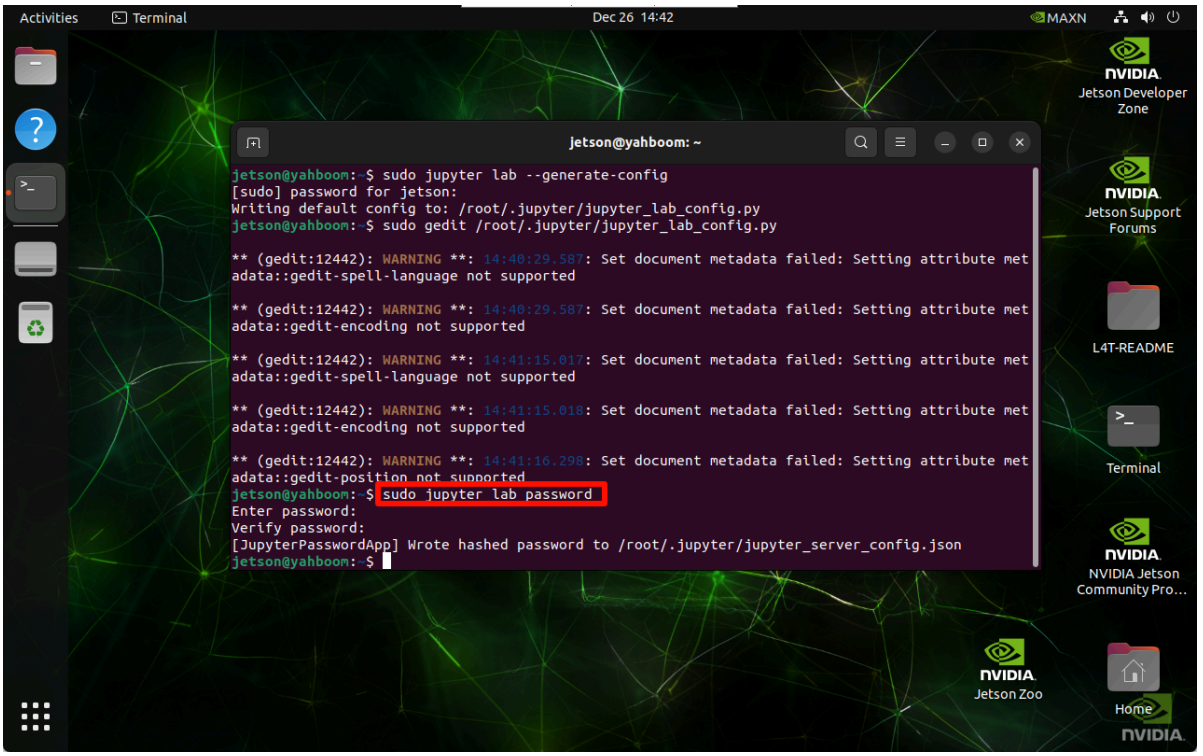


3.2. Configure the Access Password

Enter the command to set the password in the terminal; it needs to be entered twice, and the entered password is not displayed!

```
sudo jupyter lab password
```

Location of the automatically generated configuration file: [JupyterPasswordApp] Wrote hashed password to /root/.jupyter/jupyter_server_config.json



3.3. Service Auto-Start at Boot

3.3.1. Edit the Service File

```
sudo gedit /etc/systemd/system/jupyterlab.service
```

Added content: after adding, click save and close the file

```
[Unit]
Description=jupyterlab
After=network.target
[Service]
Type=simple
ExecStart=/usr/local/bin/jupyter-lab
config=/root/.jupyter/jupyter_lab_config.py --no-browser
User=root
Group=root
WorkingDirectory=/home/jetson/
Restart=always
RestartSec=10
[Install]
WantedBy=multi-user.target
```

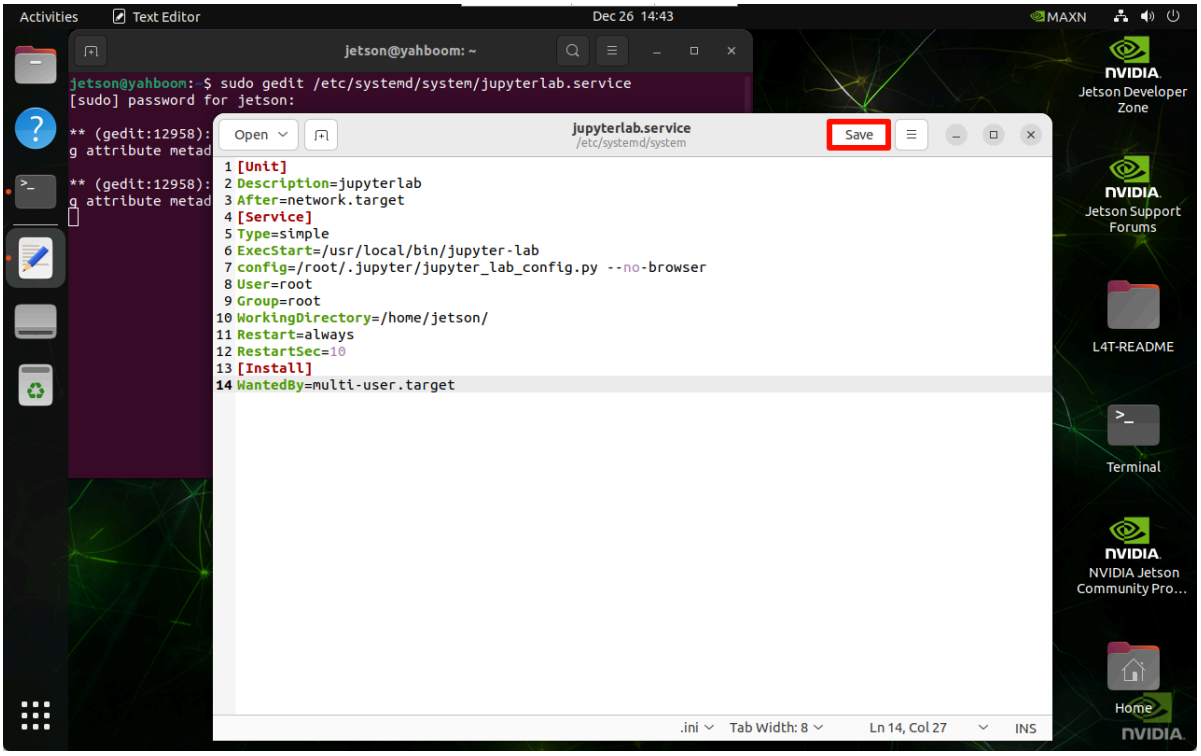
root: the system user name

ExecStart: the command to start Jupyter lab; change it to the JupyterLab installation path

config: change it to the JupyterLab configuration file path

WorkingDirectory: the working directory opened when Jupyter-lab is started; it can be changed as needed (changing it to the user directory is recommended)

```
Check the Jupyter-lab installation path: which jupyter-lab
Configuration file path: refer to the path of the configuration file generated above
```



3.3.2. Set Up the Auto-Start Service

Service Auto-Start at Boot

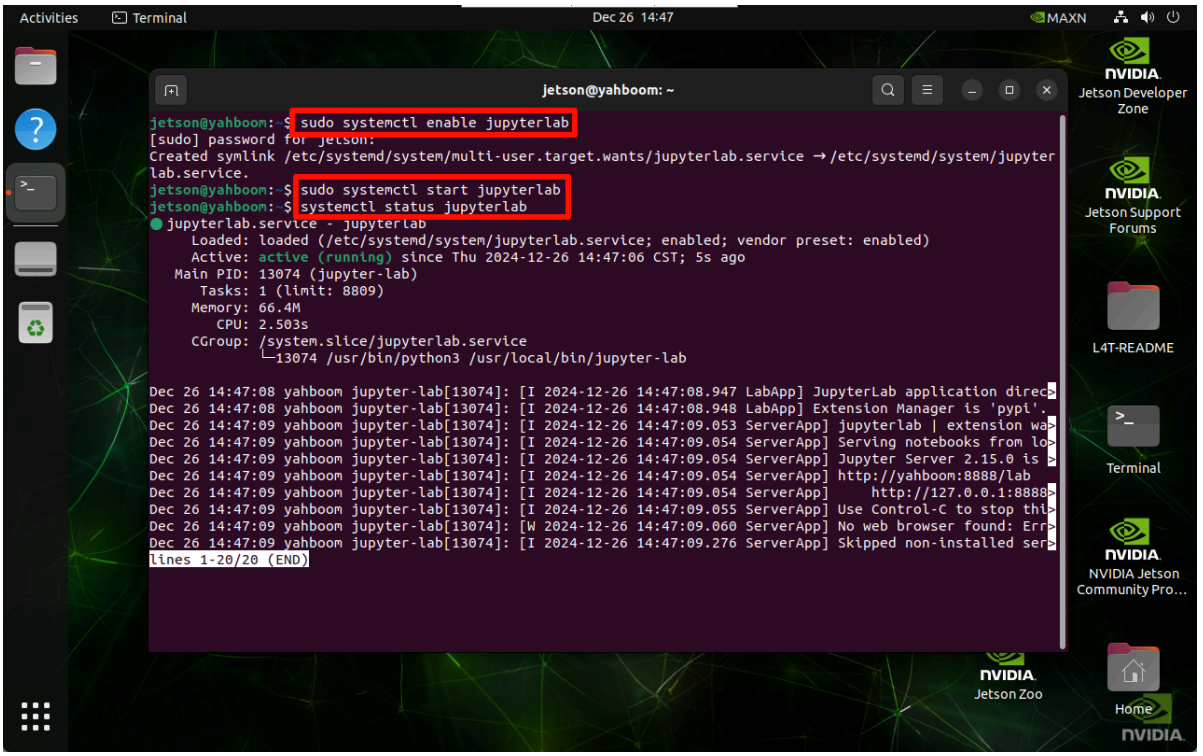
```
sudo systemctl enable jupyterlab
# Disable auto-start at boot systemctl disable jupyterlab
```

Start the Service

```
sudo systemctl start jupyterlab
# Stop the service sudo systemctl stop jupyterlab
```

Check the Service Status

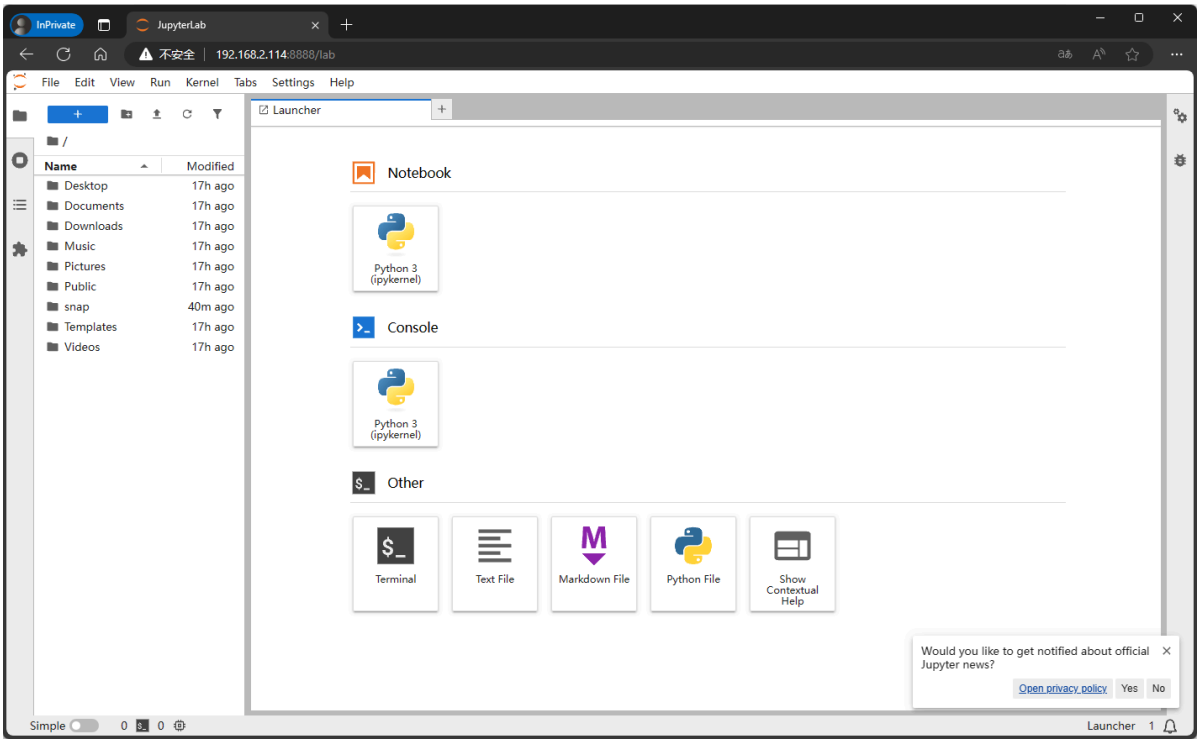
```
systemctl status jupyterlab
```



Verify Auto-Start at Boot

After rebooting the system, use a device on the same LAN to access the mainboard IP:8888 according to the system IP.

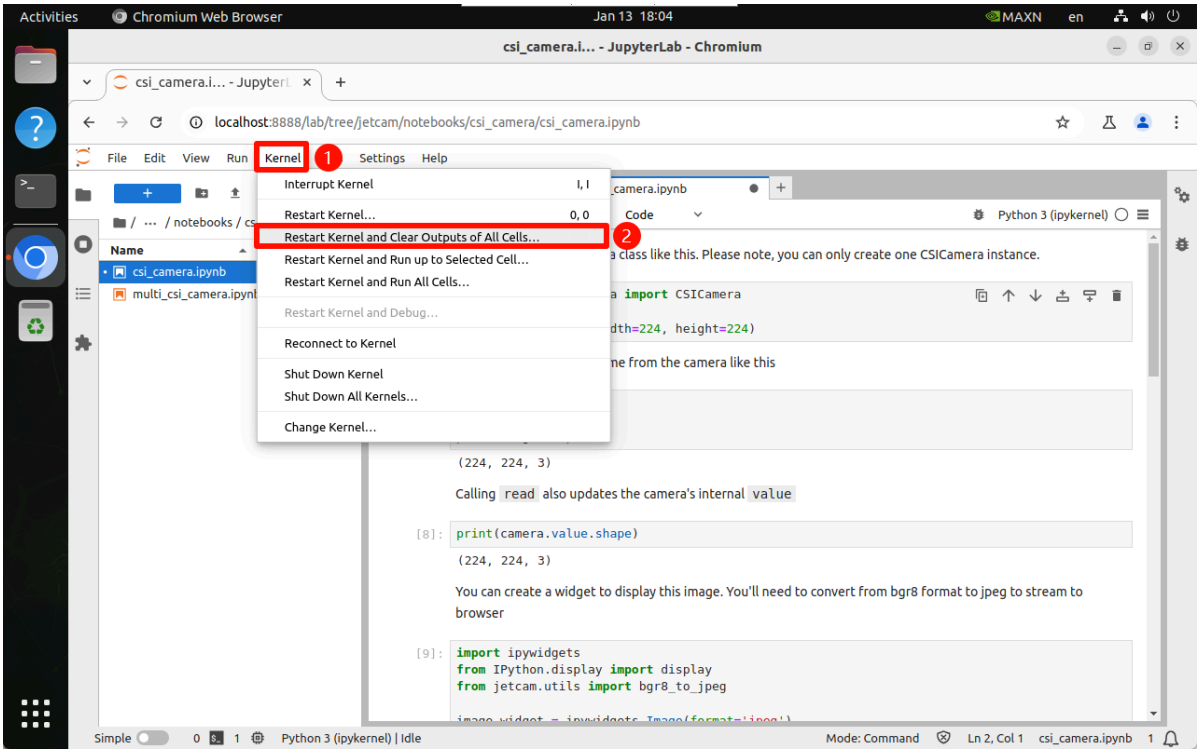
```
A password is required for the first access; the password is the information set in the previous steps;
At the time of the screenshot, the mainboard IP is 192.168.2.114, so devices on the same LAN can access 192.168.2.114:8888
```



4. Using Jupyter Lab

4.1. Kernel

It is recommended to restart the kernel and clear the output information of all cells each time a program is run or when a program is abnormal:

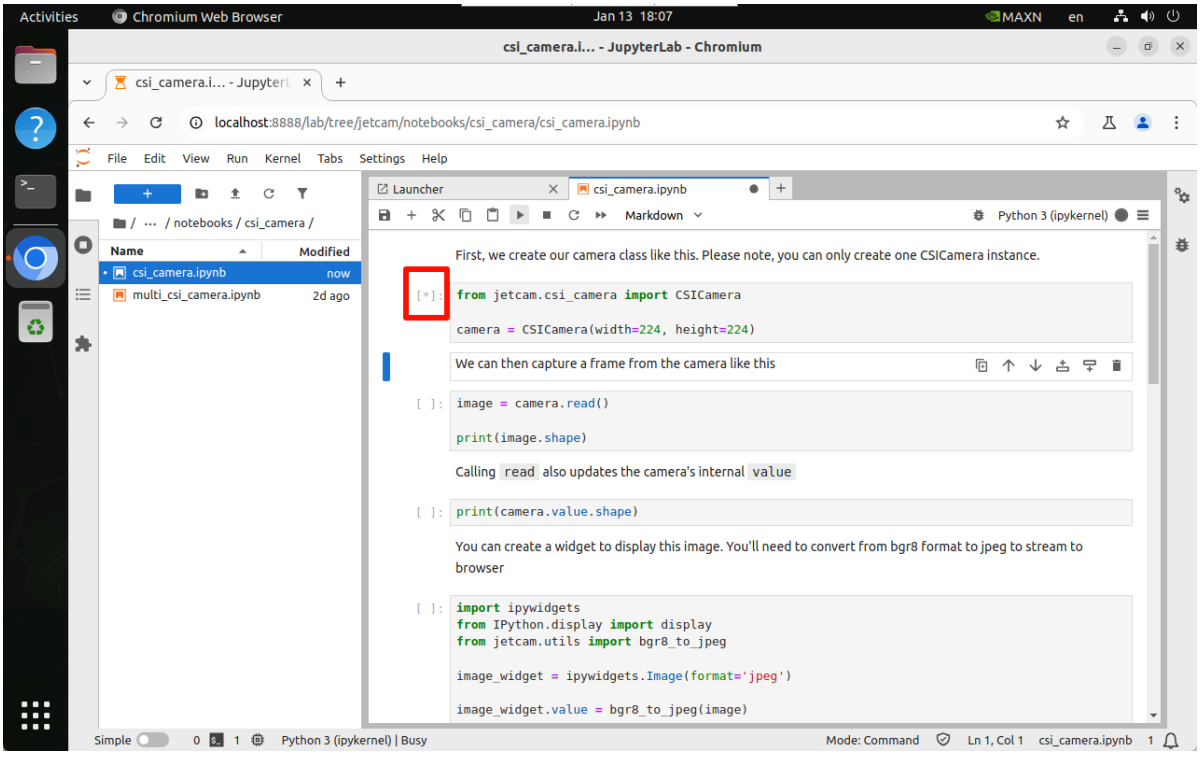


4.2. Run the Program

Open the program file to be run through Jupyter Lab, and run the cells in sequence from top to bottom when running the program:

4.2.1. Running

[*] displayed at the upper left of the cell means it is running:



4.2.2. Running Completed

[Number] displayed at the upper left of the cell indicates the run sequence count: for example, [1] → the program ran the code of that cell on the first run

